Public discourse and political discourse around the Endangered Species Act (ESA): How public opinion, media framing, and institutional decision making interact to shape conservation policy outcomes.

Agnes Namyalo

Boise State University

November 18, 2025

Abstract

Wildlife conservation remains a critical global issue due to increasing biodiversity loss and human pressures on natural habitats. Despite recognition of the importance of wildlife conservation, many conservation activities are politically contentious. Individual values, deeply held beliefs regarding appropriate behaviors and desirable outcomes, may play a role in these policy conflicts. At the same time, increasing polarization and partisanship surrounding environmental issues suggest that wildlife conservation may be a proxy for broader political battles. The role of the media, as a conduit of information between people and their elected officials in contributing to or diffusing these policy conflicts is unknown. This paper reviews a combination of quantitative and qualitative methods offering an investigation on public attitudes and policy controversies surrounding media narratives that amplify controversy despite consistently strong public support about the Endangered Species Act (ESA). We then explore emerging computational methods, specifically Natural Language Processing (NLP) techniques, for topic modeling, to evaluate how advanced text-analysis tools can uncover patterns in media framing about wildlife conservation.

1 Introduction

Wildlife conservation remains a persistent global challenge that integrates ecological science, social values, and political choice [31]. The concerns and objectives of wildlife conservation have evolved over the past century from single-species management for production to ecosystem management for biodiversity conservation. At the same time, the importance of individuals, communities, and societies in shaping conservation outcomes has been increasingly recognized. While the participants in conservation and their goals have changed, the institutions responsible for managing wildlife been less dynamic. This study revisits foundational principles and evolving perspectives that have shaped conservation thought across North America. Through a review of the literature, it highlights the intersection between policy and ecology, and underscores the growing recognition that conservation outcomes are inseparable from human governance and community engagement.

Historically, U.S. conservation has been guided by the North American Model of Wildlife Conservation (NAMWC), which provides one of the most enduring conceptual anchors for conservation policy. As shown by John F. Organ et al. [33], the model conceives of wildlife as a public trust resource, held by governments on behalf of citizens and managed for collective benefit . There are seven guiding principles to the NAMWC, including public ownership, scientific management, and regulated use developed through conceptual synthesis and historical analysis of conservation policy [13].The model has provided an institutional foundation for modern conservation practice. Yet as social values and environmental pressures evolve, the model faces renewed scrutiny. In particular, there is an increasing influence of politics in decisions about wildlife management. Instead of relying mainly on science, choices are often shaped by political views, personal beliefs, or special interests. This can lead to decisions that don't always align with science, which is intended as the foundation for wildlife policy [33, 2].One of the most important drivers behind such decisions is the influence of social values.

One critical shift comes from recognizing that conservation decisions cannot be separated from social values. Manfredo et al. [22] argue that integrating human dimensions (e.g., values, perceptions, and norms) is essential to making wildlife policy effective and legitimate. Conservation demands that we not only protect species, but also navigate how people relate to them and to each other in changing landscapes. Moreover, recent work by Artelle et al. [2] reveals that conservation decisions, despite being framed as "science-based," are frequently driven by political values and institutional priorities. In practice, the tension between science and politics manifests most visibly in wildlife policy conflicts surrounding species such as wolves and grizzly bears, where ecological management becomes a proxy for cultural and ideological divisions.

Over time, the media has become a central actor in shaping how wildlife policy debates reach both the public and policymakers. Historically, mainstream national newspapers and broadcast outlets served as gatekeepers that filtered political information and provided a relatively consistent framing of environmental issues [11]. However, the rise of digital media, partisan news ecosystems, and social platforms has fragmented information flows, amplifying ideological conflict and selective interpretations of the ESA [5, 36]. This fragmented media not only shapes how the public perceives risks and responsibilities but also influences how policymakers interpret public preferences [30]. As a result, the media has become a key space where politicians, interest groups, and advocacy organizations argue over how to interpret the ESA. This often widens the gap between the strong public support for the law and the political stories that shape legislative decisions.

Within this context, I focus on the growing misalignment between public sentiment and how wildlife policy issues are framed and addressed in both the media and legislation. For instance, although national surveys consistently show that the public strongly supports the Act, both politicians and the media regularly portray the ESA as a lightning rod for ideological conflicts between interest groups and different demographics [8]. Despite widespread public support, there has been a sharp rise in proposed rollbacks to the ESA. These patterns suggest that policymakers and the media are decoupled from public views.

This paper synthesizes foundational and contemporary perspectives on wildlife conservation governance to examine the growing misalignment between public preferences and policy outcomes. While shifts in social values—such as rising mutualism and enduring domination orientations offer one explanation for this gap, alternative perspectives point to broader forces such as increasing political polarization and contested institutional roles. Section 3 reviews how these value-based and political dynamics intersect, highlighting the uncertain yet frequently invoked role of media in shaping public perceptions and policy debates. Methodologically, it reviews both quantitative and qualitative methods used to assess ESA support in Section 4. It then examines emerging computational methods specifically, large language models-based topic modeling, as advanced tools in Section 5.

2 Social values to Wildlife conservation

Understanding social values is important to shaping effective wildlife conservation strategies. Schwartz and Bilsky define values as core motivational goals that influence patterns of behavior among people and society [37] which direct behavior across many situations [19]. Manifredo et al. [20] argues why social values are useful in wildlife conservation, although other social variables could (and in certain cases would) be useful to consider in informing wildlife conservation decisions.

Although values are often assumed to influence how people think about wildlife issues like species protection and habitat management, the empirical evidence is more mixed. Research shows that basic values alone do not strongly predict behavior or specific attitudes [16], which is why scholars such as Manfredo and Teel have introduced the concept of value orientations as a more useful framework for understanding how social values relate to wildlife conservation. [25, 21, 40]. These values help explain why individuals and communities respond differently to conservation actions. Moreover, as highlighted by the Intergovernmental

Science-Policy Platform on Biodiversity and Ecosystem Services (IPBES) [17, 12], understanding shared public values ensures that diverse voices are represented in wildlife governance, not only those of the most vocal or directly affected stakeholders.

Building on this foundation, Manfredo and colleagues developed the concept of Wildlife Value Orientations (WVOs) to categorize common patterns of values that shape how people relate to wildlife. These WVOs are shown between two orientations: mutualism, where wildlife are seen as part of the social community deserving moral consideration, and domination, where wildlife are viewed mainly in terms of human use and control. Understanding these value orientations helps explain the public attitudes and conflicts surrounding wildlife conservation issues.

2.1 Mutualism Values

Manfredo et al. [24] describe mutualism as a value orientation where people perceive wildlife as part of their social community, deserving of moral care and protection similar to humans. According to their study, individuals who hold mutualistic values believe that wildlife should have rights and legal protections, and that coexistence with animals is a shared social responsibility. This reflects a moral shift in society, where people increasingly see wildlife not as resources but as living beings to be respected.

Jacobs et al. [18] further argue that these mutualistic beliefs are spreading globally, especially among younger and urban populations. Their cross-national comparison of student populations across seven countries found that modernization, education, and urban lifestyles promote stronger mutualism orientations. These changes are associated with increased public concern for animal welfare and support for policies emphasizing coexistence.

Similarly, Manfredo et al. [20] emphasize that mutualism represents a broader sociocultural shift in conservation. They note that as societies modernize, individuals adopt more empathetic views toward wildlife. Teel and Manfredo [39] also support this, showing that mutualism influences how people respond to conservation policies and wildlife-related conflicts, generally favoring management approaches.

2.2 Domination Values

In contrast, domination values reflect the belief that humans are superior to wildlife and that animals exist primarily for human use or control. Manfredo et al. [26] describe domination-oriented individuals as those who prioritize human needs over ecological concerns and support the management or removal of animals that threaten property, safety, or economic activities. These individuals often see wildlife as something to be controlled or hunted as a form of conservation.

This control means actions like hunting, trapping, or removing animals that are seen as threats to people's property, safety, or businesses.

Jacobs et al. [18] found that domination values remain more prevalent in rural and agricultural regions, where human livelihoods are closely tied to land and resource use. Similarly, Manfredo et al. [20] observed that people who hold domination values tend to resist restrictive wildlife policies and prefer traditional management approaches that reinforce human authority over nature. These findings highlight how differing value systems of mutualism versus domination can shape social divisions around wildlife conservation decisions.

2.3 How Social Values Predict Public Support for Policy and Management

Several studies suggest that social values play an important role in shaping public attitudes toward wildlife policy and management. According to Manfredo et al. [24], mutualists are more likely to support conservation policies that emphasize coexistence, animal rights, and environmental protection, while those with domination values prioritize human benefit and control. Similarly, Manfredo et al. [26] [20] report that domination-oriented groups often resist conservation regulations or species protections that interfere with economic or personal freedoms.

Jacobs et al. [18] further demonstrate that cultural and geographic factors influence these orientations. Urbanization and modernization tend to increase mutualism values, while rural, resource-dependent communities maintain more domination-based beliefs. Manfredo et al. [23] add that these orientations also affect how individuals interpret scientific information and policy messages, thereby shaping public support or opposition to conservation measures.

The study by Manfredo et al. [26] goes further to explain how social trust and cultural differences influence these relationships. When government agencies or conservation organizations implement policies that seem misaligned with public values this creates social conflict. Therefore, understanding social values provides one framework for anticipating public reactions and designing policies that align with societal expectations.

From the perspective of this paper, understanding social values remains one useful framework among several for interpreting public reactions to wildlife policy. However, as the broader literature suggests, value change alone may not fully explain observed misalignments. It is also possible that institutional processes respond more slowly than shifts in public sentiment, or that broader political polarization overshadows the influence of individual values. In addition, communication between policymakers and the public may be complicated by media framing, which can amplify conflict or distort scientific and policy information. This complexity helps motivate the central question of the paper: why, despite generally strong and stable public support for conservation, do policy outcomes under the ESA often appear misaligned with societal preferences?

Figure 1 illustrates domination and mutualism orientations, highlighting differences in values, concerns, and preferred management actions related to wildlife.

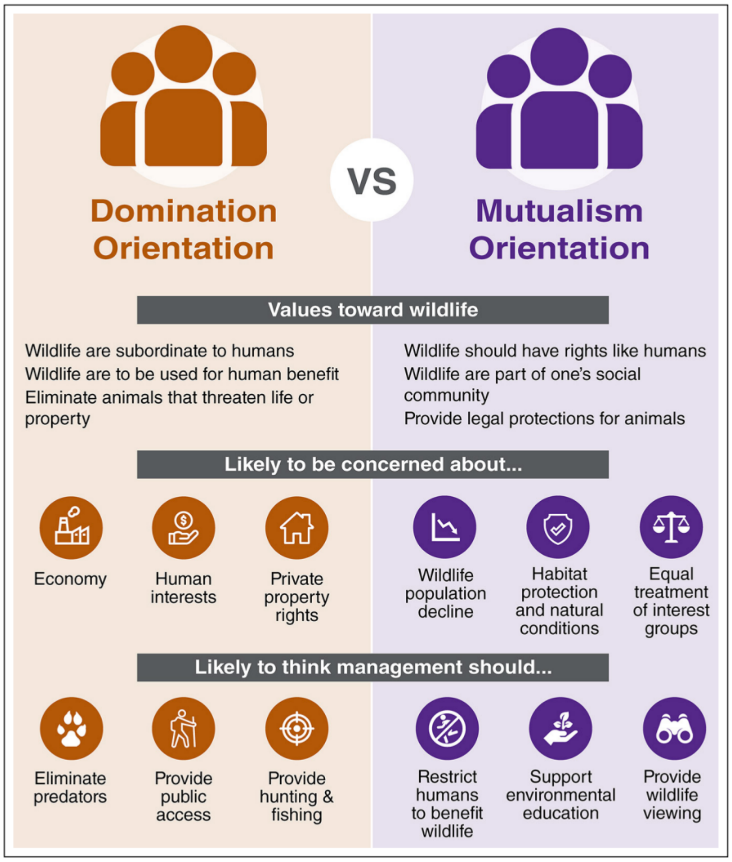

Figure 1: Comparison of Domination and Mutualism Orientations in Wildlife Values, Concerns, and Management Preferences. [20]

3 Wildlife Conservation, Misalignment, and Policy under the ESA.

For decades, wildlife conservation in the United States has been shaped by both scientific aims and political struggle.[9] Effective conservation depends on legislation designed to protect biodiversity, however political and social forces influence the extent to which the goals of such legislation are achieved [9]. In 1973 the U.S. Congress passed the the Endangered Species Act (ESA) [1] to provide for the conservation of species, where the ESA explicitly acknowledges that numerous species were "rendered extinct as a consequence of economic growth and development" and sought to provide a means of mitigating these circumstances [9],but over time its implementation and status have become entangled in political debate and institutional contestation.

According to Bruskotter and Shelby, political debates surrounding ESA reveal the persistent tension between ecological science and human interests leading to misalignment between policy implementation outcomes [7]. Nie similarly explains that the ESA has become a "lightning rod" for political and cultural battles, with wildlife management entangled in disputes over property rights, rural livelihoods, and environmental governance. In his analysis, Nie shows that endangered species debates often extend beyond biology, shaping and reflecting national political identities. The ESA, intended as a scientific conservation tool, thus becomes a stage for political negotiation and institutional power struggles [32].

Media narratives through the framing of stories covered reinforce and amplify these divisions. As noted by Hamilton et al, [15] national and local media often frame ESA controversies using conflict-based storytelling, framing the ESA as contentious. Such framing can shape how the public perceives both risk and responsibility, influencing how legislators prioritize policy responses. In Idaho, for example, wolf recovery debates have become emblematic of the state–federal divide, with news outlets alternately describing the issue as either federal overreach or local success, depending on political orientation [15].

Despite many claims in public discourse that the ESA is becoming more controversial, nationally representative surveys show consistent, strong public support for the law. Roughly four in five Americans favor the ESA across multi-decadal studies. Recent work extends this pattern through 2025 and finds support near 84%, while opposition remains low at about 12% [9, 8].

Though media coverage can influence Congressional actions, it's not always the case indeed. Media may amplify or frame issues, but Congressional action is not always a direct response to media or public opinion.

Figure 2 illustrates the conceptual flow of information and influence among media coverage, public discourse, and congressional action in U.S. wildlife policy debates.

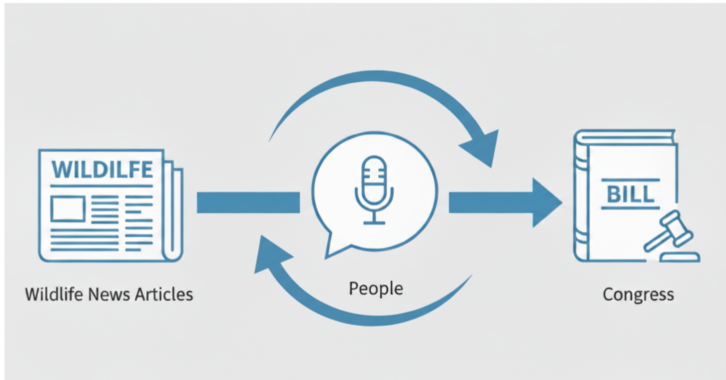

Figure 2: Interactions among wildlife news, public opinion, and Congress in shaping wildlife policy

4 Methodological Approaches in Assessing ESA Support and Policy Controversies.

Bruskotter et al. [9] and related scholars have employed a combination of quantitative and qualitative methods to investigate public attitudes and policy controversies surrounding the ESA. Together, they provide a comprehensive, multi-level understanding of how public opinion, media framing, and institutional decision-making interact to shape conservation policy outcomes.

4.1 Statistical methods, including regression analysis and tests like ANOVA and Fisher's exact test: Bruskotter et al

Bruskotter et al.[9] employed a mixed-methods approach, which quantified the stability of public support for the ESA and explored whether controversies around specific species affected attitudes, ultimately providing insight into the sociopolitical dynamics influencing ESA perceptions.

- **Dataset used**: The authors analyzed data from multiple nationally surveys and polls conducted over two decades. These surveys measure public attitudes and support for the ESA.
- **Methodology**: The authors use statistical methods, including regression analysis and tests like ANOVA and Fisher's exact test, to examine support across different groups, regions, and political ideologies. They explore correlations between support for the ESA and variables such as political ideology, identification with interest groups, and regional differences.
- **Results and findings**: The authors compare survey results across multiple years (1996–2015) to assess trends and stability in public opinions about

the ESA. Their results show that the ESA support varied from a low of 79% (±3 points, 95% CI) in 2014 to a high of 90% (±4 points, 95% CI) in 2015; opposition varied from a high of 16% (±4 points, 95% CI) in 1996 to a low of 7% (±4 points, 95% CI) in 2015 (see Figure 3). However, the margins of error surrounding the estimates of opposition in the oldest (1996) and most recent (2015) studies do not overlap—suggesting that opposition to the ESA has decreased over the past two decades [9].

Figure 3 presents survey-based frequencies of public support and opposition for the ESA across several years and sample sizes, highlighting consistently strong majorities in favor of conservation and relatively low levels of opposition. These results align with broader trends illustrated in Conservation Letters by Bruskotter et al. [9], emphasizing that average support for the ESA remains high while opposition persists at much lower rates throughout the period studied.

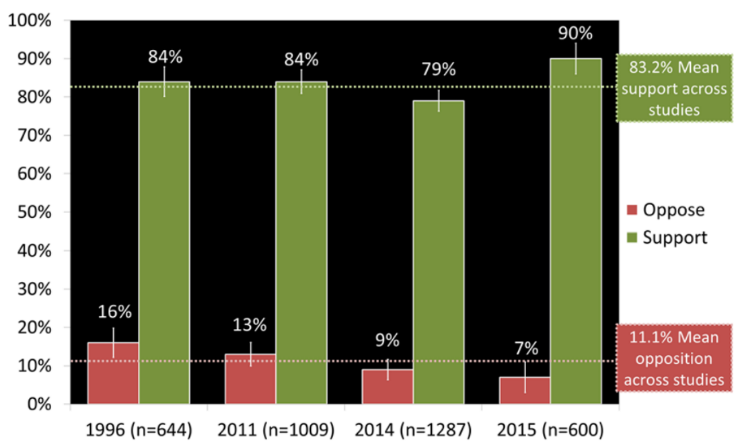

Figure 3: Americans' support for and opposition to the U.S. ESA of 1973 (1996–2015) [9]

4.2 Content Analysis approach: Vucetich et al

Vucetich et al employed a longitudinal meta-analytic approach. This comprehensive approach allows assessment of long-term public attitudes and societal factors influencing conservation policy support.

- **Dataset used**: The authors compiled data from six nationally representative surveys conducted between 1996 and 2025, measuring public attitudes and support for the ESA across time and demographic groups.
- **Methodology**: The authors employed weighted descriptive statistical analyses, including frequency distributions, confidence intervals, and subgroup comparisons by political ideology, geographic region (urban-rural), and interest group affiliations. They analyzed trends to assess stability or

changes in public support using cross-sectional and longitudinal survey data without employing predictive modeling.

- **Results and findings**: The study found consistent, high public support at about 84% (Figure 4A) and low opposition averaging about 12% (Figure 4B) to the ESA over three decades (1996–2025). The authors contextualized these quantitative results within ongoing legislative and political dynamics influencing ESA policy controversies.

Figure 4 shows trends in public support (panel A) and opposition (panel B) for ESA policy over time, with error bars indicating the variation in survey responses. These patterns are consistent with prior syntheses of ESA attitudes, such as those summarized by Bruskotter et al.[43], where strong support for conservation persists despite fluctuations in opposition rates.

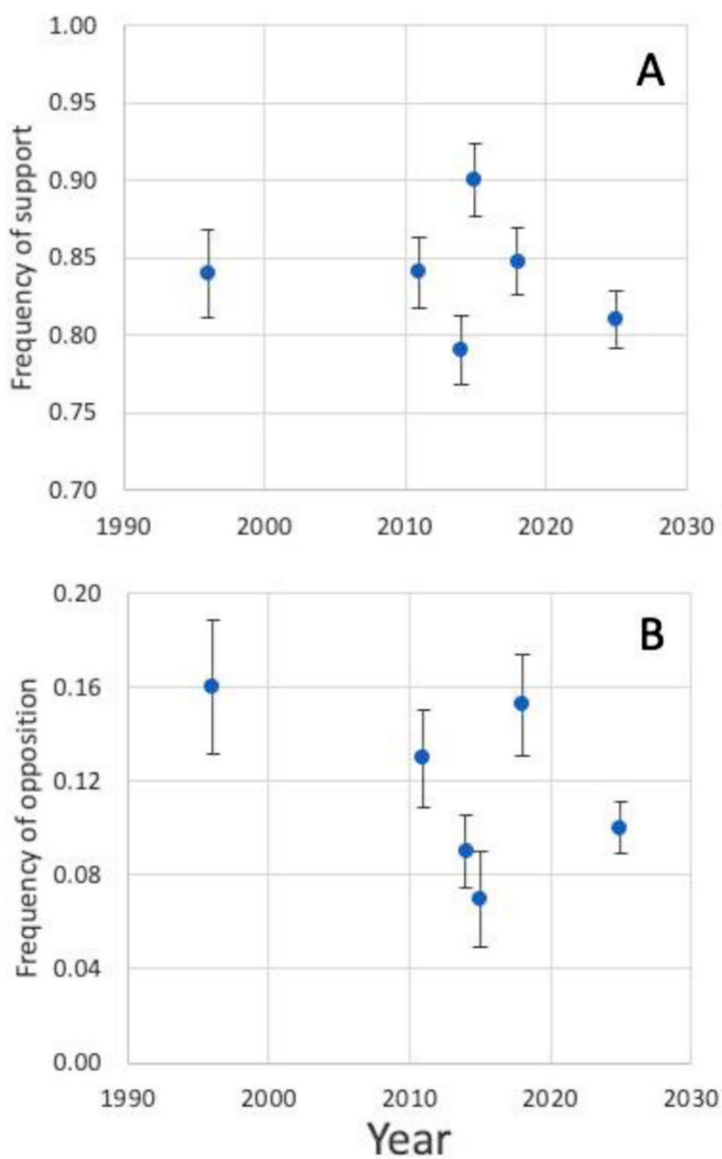

Figure 4: Support for and opposition against the US Endangered Species Act for six studies conducted over a 29-year period [43]

4.3 Common methodological patterns

Both Bruskotter et al. [9] and Vucetich et al.[43] synthesize data from multiple nationally representative surveys conducted over decades to capture trends and demographic variations in ESA support. They both apply descriptive and inferential statistical analyses, including weighting to adjust for sampling biases, subgroup comparisons by political ideology and interest group affiliation, and robust statistical testing to evaluate stability and shifts in public opinion. This reliance on large-scale, weighted survey data enables them to quantify public attitudes across time and diverse populations, providing a consistent empirical foundation for understanding the social dynamics influencing ESA policy controversies and support.

5 Large Language Models-Based Topic Modeling approach

In the realm of public policy research, especially studies of the ESA, scholars have typically utilized a blend of quantitative survey analysis (e.g., Bruskotter et al [10]), qualitative content analysis, and classical topic modeling to trace interactions among public opinion, media framing, and legislative actions. While the survey and content analysis methods offer robust insight into sociopolitical trends, they fall short in identifying emergent topics in unstructured datasets such as news articles and congressional documents. This requires the use of classical topic modeling methods such as Latent Dirichlet Allocation (LDA), Structural Topic Modelling (STM), and BERTopic.

To systematically analyze large and complex text corpora such as news articles and congressional documents, researchers rely on topic modeling approaches. These methods use statistical and computational techniques to extract the dominant topics and recurring patterns of the unstructured text. Classical models, such as Latent Dirichlet Allocation (LDA) [4], Structural Topic Modeling (STM) [35], and BERTopic [14], typically operate by evaluating word co-occurrence patterns, which allows them to group related words and documents together in a way that reflects underlying semantic structures. More recently, Large language models (LLMs) have added new powerful tools for analyzing text. They can understand the meaning, context, and much details within large sets of documents much better than older methods [6]. This makes it easier to find important themes and changes in how people talk about topics over time. Below, I review both classical and LLM-driven modeling approaches, describing their strengths, weaknesses, underlying mathematical structure and appropriate applications.

Building on this need, the study by Mu and coauthors [29] compares traditional topic modeling approaches such as LDA, STM, and BERTopic and large language models (LLMs)-based topic extraction. The paper demonstrates how LLMs, when guided through structured prompting, can outperform classical models in generating semantically coherent, interpretable, and context-sensitive topics across diverse corpora .Unlike statistical models that rely on word co-occurrence probabilities and require manual labeling of topics, The authors [3] [44] argue that traditional approaches do not perform well on handling unseen documents without a complete re-run of the model, which consequently makes it less efficient for dynamic datasets that are frequently updated (e.g., a dynamic Twitter corpus).In contrast, LLMs overcome these limitations by leveraging contextual embeddings and zero-shot generalization capabilities, allowing them to dynamically interpret new documents without full retraining. In addition, the study emphasises that [29] generative transformer-based large language models (LLMs) [42], such as GPT [6] and LLaMA [41], have an advantage of obtaining significant attention for their proficiency in understanding and generating human-like languages.

Through experiments on multiple datasets, [29] as listed below,LLMs bring transformative potential to topic modeling.Given that LLMs have strong capabilities for zero-shot text summarization comparable to human annotators [46], they can leverage their inherent understanding of language nuances to extract or generate topics. Their ability to comprehend context, thematic undertones, and linguistic subtleties has been shown across various Natural Language Processing (NLP) tasks [45, 38]. This adaptability makes LLM-based topic discovery especially effective for large-scale, evolving policy corpora, enabling real-time detection of emerging narratives and issue framings.

The authors [29] structure their methodology around three experimental components:

- Prompting LLMs to generate topic labels directly from corpora.
- Clustering documents based on semantic similarity.
- Evaluating topic coherence through both automated and human-centered assessments.

Figure 5 shows two final topics generated by the LLMs, namely “Technology & Computers” (20NG) and “Trust & Mistrust” (Vaccine dataset), highlighting the distinct semantic patterns captured across datasets [29].

Figure 5: Two examples of final topics summarised by LLMs.[29]

Results indicate that LLM-based topic modeling produces more interpretable topic summaries, higher coherence scores, and enhanced adaptability to dynamic datasets. Furthermore, human evaluators consistently rated LLM-generated topics as clearer and more semantically accurate. Nonetheless, Mu et al [29] acknowledge challenges, including sensitivity to prompt design, computational cost, and occasional instability when processing highly technical or domain-specific language.

To better understand how these approaches differ in their capacity to model and interpret large-scale policy texts, it is useful to examine their underlying mathematical structures. This shows how each method operates under distinct mathematical assumptions and computational mechanisms that shape its interpretability and flexibility when applied to dynamic policy datasets.

5.1 Structural Topic Model (STM)

The Structural Topic Model (STM) extends the Latent Dirichlet Allocation (LDA) framework by incorporating document-level covariates into both the *topic prevalence* and *topic content* components. This enables the model to capture how external factors (e.g., author, time, or ideology) influence both the proportion of topics in a document and the words used to express them [35].

Specifically, STM introduces two key components:

1. **Topic prevalence** is modeled as a function of document-level covariates.
2. **Topic content** can also vary with covariates, such as ideological bias or publication source.

The core generative process of STM is defined as:

$$\theta_d \sim \text{LogisticNormal}(X_d\gamma, \Sigma) \tag{1}$$
$$z_{dn} \sim \text{Multinomial}(\theta_d) \tag{2}$$
$$w_{dn} \sim \text{Multinomial}(\beta_{z_{dn}}) \tag{3}$$

Here:

- X_d = document-level covariates (e.g., date, author, or region),
- γ = regression coefficients capturing covariate effects on topic proportions,
- Σ = covariance matrix representing topic correlations,
- θ_d = document-topic proportions, now dependent on covariates,
- $\beta_{z_{dn}}$ = topic-word distributions, potentially varying by covariate groups.

5.2 BERTopic

In contrast to the probabilistic methods, BERTopic leverages deep neural embeddings to capture contextual similarity between texts.

BERTopic leverages transformer-based embeddings to capture semantic relationships, followed by dimensionality reduction and clustering [27, 28, 14].

Specifically, Uniform Manifold Approximation and Projection (UMAP) is applied to reduce the high-dimensional BERT embeddings into a lower-dimensional space while preserving their local and global structure, thereby facilitating efficient clustering. Hierarchical Density-Based Spatial Clustering of Applications with Noise (HDBSCAN) then groups documents into clusters of varying density, automatically identifying outliers and adapting to complex topic distributions. Finally, within each cluster, class-based Term Frequency–Inverse Document Frequency (c-TFIDF) extracts the most representative words, yielding interpretable topic representations.

$$E = \mathrm{BERT}(D) \tag{4}$$

$$E' = \mathrm{UMAP}(E) \tag{5}$$

$$C = \mathrm{HDBSCAN}(E') \tag{6}$$

$$\mathrm{Topic}_k = \text{c-TFIDF}(C_k) \tag{7}$$

Here, E denotes contextual embeddings, E' reduced representations via UMAP, and C cluster assignments. Topics are extracted from each cluster using class-based TF-IDF (c-TFIDF). This pipeline allows BERTopic to handle high-dimensional, unstructured policy texts with improved semantic granularity.

5.3 LLM-Based Topic Extraction: GPT and LLaMA

Generative transformer-based large language models (LLMs) such as GPT [34] and LLaMA [41] redefine topic discovery by replacing probabilistic inference with prompt-driven contextual understanding. Each transformer layer computes attention as:

$$\mathrm{Attention}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V \tag{8}$$

where Q, K, and V are the query, key, and value matrices, respectively. Through stacked self-attention layers, LLMs dynamically learn dependencies and thematic coherence across long text sequences.

In the context of environmental and wildlife policy research, these methodological innovations hold significant potential. As scholars seek to trace the evolution of discourse around the ESA, wildlife reintroduction, and climate governance, LLM-based topic modeling provides an avenue to explore how media, and public narratives co-evolve. By merging interpretability and scalability, the LLM approach introduced by Mu et al. [29] represents a promising direction for synthesizing complex political and ecological discourses across time.

6 Conclusion

While this synthesis has shown persistent misalignments between public sentiment, media narratives, and legislative action under ESA, it also reveals opportunities for advancing our understanding of these dynamics through new computational approaches. The next phase of my research will directly investigate the relationship between media and the public: Who leads, and who follows, in shaping wildlife conservation discourse in the U.S.? Specifically, I propose to use large language models (LLMs) alongside traditional topic modeling (such as STM, and BERTopic) to compare how each method detects shifts in topics, framing, and agenda setting across news articles and congressional documents around controversial conservation issues like the ESA.I aim to clarify whether and when the media is driving public opinion, or instead responding to it. This approach not only addresses methodological scalability but will also help determine the advantages of LLMs over classical techniques for topic modeling.

References

[1] Endangered species act of 1973. Pub. L. No. 93-205, 87 Stat. 884, 1973. U.S. Congress.

[2] Kyle A Artelle, John D Reynolds, Adrian Treves, James C Walsh, Paul C Paquet, and Chris T Darimont. Hallmarks of science missing from north american wildlife management. *Science Advances*, 4(3):eaao0167, 2018.

[3] David M Blei and John D Lafferty. Dynamic topic models. In *Proceedings of the 23rd international conference on Machine learning*, pages 113–120, 2006.

[4] David M. Blei, Andrew Y. Ng, and Michael I. Jordan. Latent dirichlet allocation. *Journal of Machine Learning Research*, 3:993–1022, 2003.

[5] Maxwell T. Boykoff and Jules M. Boykoff. Environmental citizenship and the politics of science. *Political Geography*, 26(3):470–491, 2007.

[6] Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, et al. Language models are few-shot learners. *Advances in neural information processing systems*, 33:1877–1901, 2020.

[7] Jeremy T Bruskotter and Bo Shelby. The endangered species act and its role in conservation politics. *Conservation Biology*, 24(1):1–8, 2010.

[8] Jeremy T. Bruskotter, John A. Vucetich, Alia M. Dietsch, and Kevin M. Slagle. Public support for the endangered species act remains strong despite political polarization. *Frontiers in Ecology and the Environment*, 21(2):94–102, 2023.

[9] Jeremy T. Bruskotter, John A. Vucetich, Alia M. Dietsch, Kevin M. Slagle, Jeremy S. Brooks, Michael P. Nelson, and Sara J. Bruskotter. Support for the u.s. endangered species act over time and space: Controversial species do not weaken public support for protective legislation. *Conservation Letters*, 11(6):e12595, 2018.

[10] Jeremy T. Bruskotter, John A. Vucetich, et al. Support for the u.s. endangered species act over time and space: Controversial species do not weaken public support for protective legislation. *Conservation Letters*, 11(6), 2018.

[11] Robert Cox and Phaedra Pezzullo. *Environmental Communication and the Public Sphere*. Sage, 2016.

[12] Sandra Díaz, Sebsebe Demissew, Julia Carabias, Carlos Joly, Mark Lonsdale, Neville Ash, Anne Larigauderie, Jay Ram Adhikari, Salvatore Arico, András Báldi, et al. The drivers of change in biodiversity and ecosystem services. *Science*, 348(6242):124–129, 2015.

[13] Valerius Geist, Shane P Mahoney, and John F Organ. The north american model of wildlife conservation. *Wildlife Society Bulletin*, 29(2):407–450, 2001.

[14] Maarten Grootendorst. Bertopic: Neural topic modeling with a class-based tf-idf procedure. 2022.

[15] Lawrence C Hamilton, Joel Hartter, and Andrew Bell. Framing wildlife policy in the news: How media narratives shape public perception of conservation. *Environmental Communication*, 14(2):234–249, 2020.

[16] Thomas A. Heberlein. The roles of values, attitudes, and norms in environmental behavior. *Human Ecology Review*, 19(2):121–130, 2012.

[17] IPBES. *The IPBES regional assessment report on biodiversity and ecosystem services for the Americas*. Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services, 2018.

[18] Maarten H. Jacobs, Jerry J. Vaske, Tara L. Teel, et al. Exploring cultural differences in wildlife value orientations using student samples in seven nations. *Biodiversity and Conservation*, 2022.

[19] Michael J Manfredo. Who cares about wildlife? In *Who cares about wildlife? Social science concepts for exploring human-wildlife relationships and conservation issues*, pages 1–27. Springer, 2008.

[20] Michael J Manfredo, Richard EW Berl, Tara L Teel, and Jeremy T Bruskotter. Bringing social values to wildlife conservation decisions. *Frontiers in Ecology and the Environment*, 19(6):355–362, 2021.

[21] Michael J. Manfredo, Jeremy T. Bruskotter, Tara L. Teel, David Fulton, Shalom Schwartz, Robert Arlinghaus, Shigehiro Oishi, Ayse K. Uskul, Kent Redford, Shinobu Kitayama, and L. Sullivan. The origins of human values for nature: A systems perspective. *Proceedings of the National Academy of Sciences*, 113(6):1465–1472, 2016.

[22] Michael J. Manfredo, Jeremy T. Bruskotter, Jerry J. Vaske, and Andreas Rechkemmer. Bringing social values to wildlife conservation decision making. *Frontiers in Ecology and the Environment*, 18(7):355–362, 2020.

[23] Michael J. Manfredo et al. Social values and biodiversity conservation in a dynamic world. *Biological Conservation*, 201:325–332, 2016.

[24] Michael J. Manfredo, Tara L. Teel, Richard E.W. Berl, Jeremy T. Bruskotter, et al. The changing sociocultural context of wildlife conservation. *U.S. Geological Survey*, 2020.

[25] Michael J. Manfredo, Tara L. Teel, and Alan D. Bright. Values, beliefs, and norms that drive conservation efforts. *Human Dimensions of Wildlife*, 14(1):1–9, 2009.

[26] Michael J Manfredo, Tara L Teel, Leeann Sullivan, and Alia M Dietsch. Values, trust, and cultural backlash in conservation governance: The case of wildlife management in the united states. *Biological Conservation*, 214:303–311, 2017.

[27] Leland McInnes, John Healy, and Steve Astels. hdbscan: Hierarchical density based clustering. *Journal of Open Source Software*, 2(11):205, 2017.

[28] Leland McInnes, John Healy, and James Melville. Umap: Uniform manifold approximation and projection for dimension reduction. *arXiv preprint arXiv:1802.03426*, 2018.

[29] Yida Mu, Chun Dong, Kalina Bontcheva, and Xingyi Song. Large language models offer an alternative to the traditional approach of topic modelling. *arXiv preprint arXiv:2403.16248*, 2024.

[30] Diana Mutz. The influence of media on public opinion. *Annual Review of Political Science*, 18:47–66, 2015.

[31] Martin Nie. *Drivers of Natural Resource Policy: The Politics of Wildlife, Land, and the Environment*. MIT Press, Cambridge, MA, 2003.

[32] Martin A Nie. Drivers of natural resource policy change: Political conflict and institutional adaptation in the u.s. endangered species act. *Policy Sciences*, 37(4):369–392, 2004.

[33] John F Organ, Valerius Geist, Shane P Mahoney, Steven Williams, Paul R Krausman, Gordon R Batcheller, Thomas A Decker, Robert Carmichael, Priya Nanjappa, Ronald Regan, et al. The north american model of wildlife conservation. *The wildlife society technical review*, 12(04), 2012.

[34] Long et al. Ouyang. Training language models to follow instructions with human feedback. *Advances in Neural Information Processing Systems*, 2022.

[35] Margaret E. Roberts, Brandon M. Stewart, and Dustin Tingley. Structural topic models for open-ended survey responses. *American Journal of Political Science*, 58(4):1064–1082, 2014.

[36] Dietram A. Scheufele. Science communication and political polarization. *PNAS*, 110(Supplement 3):14048–14054, 2013.

[37] Shalom H Schwartz, Naomi Struch, and Wolfgang Bilsky. Values and intergroup social motives: A study of israeli and german students. *Social Psychology Quarterly*, pages 185–198, 1990.

[38] Xiaoyu Tang, Jian Zhao, and Rui Liu. Large language models for deep thematic clustering in policy texts. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 10211–10225, 2023.

[39] Tara L. Teel and Michael J. Manfredo. Public perceptions of wildlife-associated recreationists across north america and europe. *Human Dimensions of Wildlife*, 14(6):363–379, 2009.

[40] Tara L. Teel and Michael J. Manfredo. Values, value orientations, and attitudes: Clarifying concepts and measures with the use of a wildlife value orientations framework. *Human Dimensions of Wildlife*, 14(2):82–96, 2009.

[41] Hugo Touvron, Thibaut Lavril, Gautier Izacard, Xavier Martinet, Marie-Anne Lachaux, Timothée Lacroix, Baptiste Rozière, Naman Goyal, Eric Hambro, Faisal Azhar, et al. Llama: Open and efficient foundation language models. *arXiv preprint arXiv:2302.13971*, 2023.

[42] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. *Advances in neural information processing systems*, 30, 2017.

[43] John A Vucetich, Jeremy T Bruskotter, Robyn Wilson, L Mark Elbroch, Adam Feltz, and Thomas Offer-Westort. Support for the us endangered species act is high and steady over the past three decades. *Conservation Letters*, 18(3):e13111, 2025.

[44] Chong Wang, David Blei, and David Heckerman. Continuous time dynamic topic models. *arXiv preprint arXiv:1206.3298*, 2012.

[45] Haoran Wu, Rahul Singh, and Li Zhang. Llm-based semantic topic discovery: Contextual understanding beyond co-occurrence. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 3894–3908, 2023.

[46] Wei Zhang, Ming Chen, and Xiaohui Li. Harnessing large language models for zero-shot text summarization. *Transactions of the Association for Computational Linguistics*, 12:455–471, 2024.